Shortest paths

Optimization · Level 3 · Lesson 2



In this lesson you will

- Store a road map as a Python dictionary.
- Follow and complete Dijkstra’s method.
- Explain why settling the closest place first is safe.

Key ideas

- A map is a graph
- Settle the smallest number first
- Relax: keep the smaller
- No negative roads
- Who gains from a road?

Activity: My map, settled

1. Draw (5 min): Six places and the roads between them.
2. Settle (8 min): Dijkstra on paper from your home.
3. Code (5 min): Type the map; does it agree?
4. Close a road (2 min): Who is hurt most?

Use made-up names for homes. Distances can be minutes of walking instead of kilometres.

Let’s discuss

1. Why is the straightest-looking road not always the shortest?
2. What would go wrong with a negative road length?
3. Who should decide which road gets repaired first?

Exit ticket

- The one line that relaxes a road + why the smallest goes first.